EE4-07
EE9-CS7-23
EE9-SO11

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE/EIE PART IV: MEng and ACGI

CODING THEORY

Tuesday, 27 April 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

All the questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : W. Dai
Second Marker(s) : C. Ling

© Imperial College London

EE4-07 Coding Theory

Instructions for Candidates

Answer all three questions. Each question carries 25 marks.

 © Imperial College London, 2021

1. (a) Let $\mathcal{C} \subset \mathbb{F}_3^7$ be a linear code. Its parity check matrix is given by

$$\boldsymbol{H}' = \begin{bmatrix} 1 & 2 & 0 & 2 & 1 & 2 & 0 \\ 0 & 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 & 2 & 0 & 2 \end{bmatrix}.$$

i Use Gaussian elimination to change the parity matrix into the form of $\boldsymbol{H} = [\boldsymbol{I}\ \boldsymbol{A}] \in \mathbb{F}_3^{3\times 7}$ where $\boldsymbol{I}$ is the identity matrix. [2]

ii Find the corresponding generator matrix $\boldsymbol{G} \in \mathbb{F}_3^{4\times 7}$ in the form $[\boldsymbol{B}\ \boldsymbol{I}]$. [2]

iii What is the distance of the code defined by $\boldsymbol{H}$? Justify your answer. [4]

iv Assume that a message $\boldsymbol{m}_1$ is encoded into a codeword $\boldsymbol{c}_1$ using $\boldsymbol{G}$. The codeword $\boldsymbol{c}_1$ is transmitted over a ternary symmetric channel. Let the received word be

$$\boldsymbol{y}_1 = [1\ 1\ 1\ 0\ 0\ 2\ 1]\,.$$

Compute the syndrome vector $\boldsymbol{s}_1$. Find the output of the minimum (Hamming) distance decoding, say $\hat{\boldsymbol{c}}_1$. [2]

v Assume that a message $\boldsymbol{m}_2$ is encoded into a codeword $\boldsymbol{c}_2$ using $\boldsymbol{G}$ The codeword $\boldsymbol{c}_2$ is transmitted over a ternary erasure channel. Let the received word be

$$\boldsymbol{y}_1 = [1\ 0\ ?\ ?\ 1\ 0\ 2]\,.$$

Set the question marks in $\boldsymbol{y}_2$ to zero and compute the corresponding syndrome vector $\boldsymbol{s}_2$. Find the transmitted codeword $\boldsymbol{c}_2$. [2]

(b) Let $f(x) = x^3 + x + 1 \in \mathbb{F}_3[x]$.

i Let $g_1(x) = x^2 + 1 \in \mathbb{F}_3[x]$. Find the *monic* polynomial $h_1(x) = \gcd(f(x), g_1(x)) \in \mathbb{F}_3[x]$ using Euclidean algorithm. Based on Bezout's identify, find $a_1(x), b_1(x) \in \mathbb{F}_3[x]$ such that $h_1(x) = a_1(x)f(x) + b_1(x)g_1(x)$. [3]

ii Let $g_2(x) = x^2 + 2 \in \mathbb{F}_3[x]$. Find the *monic* polynomial $h_2(x) = \gcd(f(x), g_2(x)) \in \mathbb{F}_3[x]$ using Euclidean algorithm. Based on Bezout's identify, find $a_2(x), b_2(x) \in \mathbb{F}_3[x]$ such that $h_2(x) = a_2(x)f(x) + b_2(x)g_1(x)$. [3]

(c) Consider the finite field $\mathbb{F}_{16}$ detailed below

	$\mathbb{F}_2[z]/z^4+z+1$		$\mathbb{F}_2[z]/z^4+z+1$
0	0	z^7	z^3+z+1
1	1	z^8	z^2+1
z	z	z^9	z^3+z
z^2	z^2	z^{10}	z^2+z+1
z^3	z^3	z^{11}	z^3+z^2+z
z^4	$z+1$	z^{12}	z^3+z^2+z+1
z^5	z^2+z	z^{13}	z^3+z^2+1
z^6	z^3+z^2	z^{14}	z^3+1

i Compute $\beta_1 = (z^3+z)\cdot(z^3+z^2+z) \in \mathbb{F}_{16}$. [2]

ii Compute $\beta_2 = (z^3+z^2+z)^{-1}\cdot(z^3+z^2) \in \mathbb{F}_{16}$. [2]

iii Compute the polynomial $f(x) = (x-\beta_1)\cdot(x-\beta_2) \in \mathbb{F}_{16}[x]$. Show key steps of computations. [3]

2. Consider the prime number $p = 131$ and a primitive element $g = 54$ of $\mathbb{F}_p$.

(a) Find $g^n \bmod p$ where $n = 1,\ 2,\ 4,\ 8,\ 16,\ 32,\ 64,\ 128$, respectively. Write your final answer in a table format as below.

$n =$	1	2	4	8	16	32	64	128	
$g^n \equiv$	?	?	?	?	?	?	?	?	mod p

[4]

(b) Based on the results of the above calculations, find $x = \log_g(114) \bmod p$. [1]

(c) Alice and Bob would would like to communicate using the ElGamal public key cryptography which involves the prime number p and the primitive element g. Suppose that the private keys that Alice and Bob choose are $k_{s,a} = 72$ and $k_{s,b} = 96$, respectively.

i Find the public keys that Alice and Bob will use respectively. [2]
Note: In the following sub-questions, Alice and Bob only know each other's public key (not the other person's private key).

ii Suppose that Bob needs to send Alice a plaintext message $m = 73$. He chooses a random number $t = 80$ to add randomness to his message. Find the encrypted message that Bob will use. Show key steps of your computations. [5]

iii Suppose that Alice sends Bob an encrypted message $(S_1, S_2) = (55, 50)$. Find the plaintext message from the encrypted message. Show key steps of your calculations. [5]

(d) In this sub-question, consider another prime number $p = 107$.
Let $a = 51$ and $b = 56$. Find their orders in $\mathbb{F}_p$, denoted by $\text{ord}(a)$ and $\text{ord}(b)$ respectively. Which numbers are primitive elements in $\mathbb{F}_p$? Justify your answer. [8]

3. Consider codes on $\mathbb{F}_q$. The code size (the number of codewords) is denoted by M, the length of codewords is denoted by n, and distance of the codebook is denoted by d. If the code is linear, its dimension is denoted by k.

(a) i Let $q = 3$, $n = 10$, and $d = 5$.
What does Hamming bound predict about M and k?
What does Gilbert-Varshamov bound predict about M and k?
Show key steps of your computations. [4]

ii Let $q = 11$, $n = 10$, and $d = 5$.
What does Hamming bound predict about M and k?
What does Gilbert-Varshamov bound predict about M and k?
Show key steps of your computations. [4]

(b) Consider field $\mathbb{F}_p$ with $p = 7$ and a primitive element $g = 3$.

i Let $n = 6$ and $d = 5$. What does Singleton bound predict about k? [1]

ii The above Singleton bound can be attained by a Reed-Solomon code $\mathcal{C}$. It can be defined by its parity check matrix of the form

$$\boldsymbol{H} = \begin{bmatrix} 1 & g & \cdots & g^{n-1} \\ 1 & g^2 & \cdots & g^{2(n-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & g^{n-k} & \cdots & g^{(n-k)(n-1)} \end{bmatrix}.$$

Compute the parity check matrix $\boldsymbol{H}$ of the Reed-Solomon code that attains what you computed above. This will allow you to do decoding below. [4]

iii Suppose that a codeword $\boldsymbol{x} \in \mathcal{C}$ is transmitted and the received word is given by

$$\boldsymbol{y} = \boldsymbol{x} + \boldsymbol{e} = [4\ 6\ 3\ 2\ 3\ 1].$$

Compute its syndrome vector $\boldsymbol{s}$. Find the corresponding syndrome polynomial $S(z)$. [5]

iv Using the key equation

$$E(z) = L(z) * S(z) \bmod z^{d-1},$$

find the corresponding error locator polynomial $L(z)$ and error evaluator

polynomial $E(z)$. [5]

v Using the locator polynomial $L(z)$ find the error locations. [2]